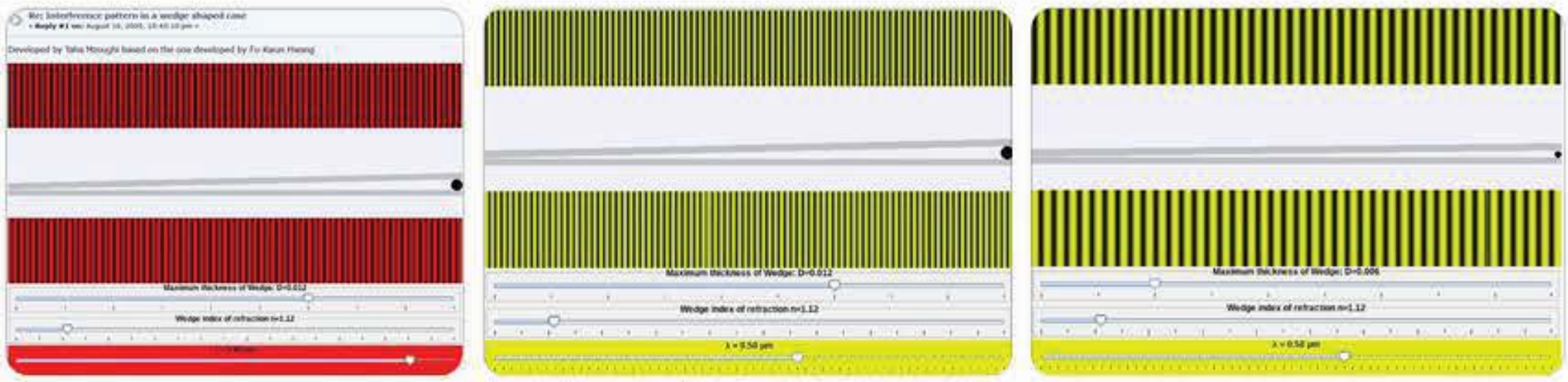


This is something I worked on many years back (the date there is 2009). Let us try reviving it. Open to other layouts, especially, to help explaining what happens. You might also would like to work on a Newton's rings version. Having a white light toggle will be nice. (We will later try a general thin film, with like this, reflection and transmission views, and soap bubble application)

Thought for 53s

Glad to revive it. Your original shows the essentials (the wedge between two plates, and the fringes seen from both sides), but not *why* the fringes appear. So I'd build the new version around the explanation, and add Newton's rings as a second tab.

Tab 1, the wedge, as a stack of aligned views, so any point can be read straight down through all of them:

- **The pattern seen in reflection,** then **a magnified side view of the wedge,** then **the pattern seen in transmission.**
- **A probe line you drag across.** In the side view, it shows the light reflected at the top and at the bottom of the gap, with the thickness t, the extra path 2nt, and which reflection gets the half-wave (π) phase change.
- **The two reflected waves at the probe,** drawn with their sum, so you can see them arrive in step (bright) or opposed (dark).
- **An intensity graph R(x) and T(x),** aligned with the patterns, showing that R + T = 1.

Tab 2, Newton's rings: a plano-convex lens on a flat plate. A side view with the probe, the face-on rings in reflection and in transmission, and the intensity against the radius.

For both, a **white-light** toggle (the coloured fringes, summed over the spectrum), a wavelength slider with its colour, the film's index and the glass's, and the math with your numbers.

On the physics, I'll use the exact multiple-reflection (Airy) formula rather than a two-beam approximation. It shows two things your students may not expect:

- **The transmitted fringes are really faint,** about 8 % contrast for an air gap between glass plates. A "stretch the contrast" option makes them easy to see.
- **The fringes vanish** when the film's index equals the glass's.

Let me check the key numbers first:

Ran 2 commands, created a file

Continue

Ran a command

The spacing is now 1.450 mm, and spacing × number of fringes is exactly the 60 mm plate. Now the wedge tab as a whole:

Read 4 files, ran 2 commands, shared files



Your 2009 wedge is revived as `po-thin-film-wedge-newton-rings`, with GitHub and MOM versions and a SIM description. It's in `sims-mobile.zip`, and the inventory lists 129 sims.

Tab 1, the wedge, is laid out to explain what happens. Every view is aligned, and a probe you drag across any of them shows the same point in all:

- **The fringes seen in reflection**, as in your original.
- **A magnified side view** of the two plates and the wire. At the probe it draws the light reflected at the top and at the bottom of the gap, the thickness t and the extra path 2nt, and says which reflection has the half-wave phase change.
- **The fringes seen in transmission**, as in your original.
- **The two reflected waves at the probe** and their sum, in step (bright) or opposed (dark), with the phase difference in wavelengths and degrees.
- **The brightness across the wedge**, R and T together, always summing to 1.

Tab 2, Newton's rings, has the same structure: a side view of the lens on the plate, the rings face-on in reflection and in transmission, the waves at the probe, and the brightness against the radius, with the dark rings marked.

In both:

- **White light:** the colours summed over the spectrum, giving Newton's colour sequence that washes out to white.
- **Controls:** the wavelength (its slider takes the colour), the film's index, and presets for air, water and an oil matching the glass. In that last case the fringes vanish.
- **The math, narration and accessibility toolbar** match your other sims.

The physics is exact: it uses the formula for all the multiple reflections, not just two beams. I checked the results against theory:

- **Dark fringes** fall exactly where $2nt = m\lambda$.
- **Your original settings** (D = 12 μm, 580 nm, 6 cm plates) give 41.4 fringes, 1.450 mm apart.
- **The first dark ring** for a 2 m lens is at 1.077 mm.

One honest detail: through glass plates, the transmitted fringes are really faint, about 8.5 % contrast, because most light gets through at any thickness. So they're shown "contrast stretched" by default, with the true contrast labelled; untick the option to see them as they really are.

Checked: it works on a 360 px phone, and two copies on one MOM page stay independent. When you're ready for the general thin film (reflection and transmission views, and the soap bubble), it can reuse this engine.

Po thin film wedge newton rings
Code · HTML

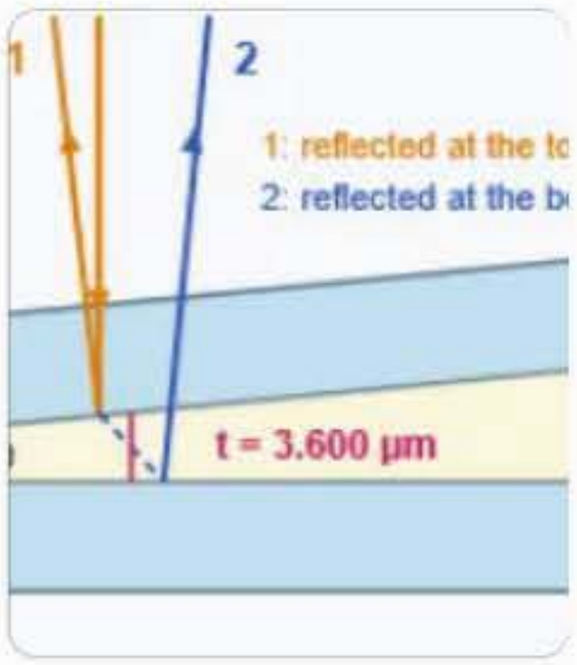
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That is good. One small addition please to help the student who need it the most. A sort of magnified view of what I pasted with more delays. Like having the thickness written in terms of wavelength. Having one cycle of the incoming, reflected and refracted waves at each of the interfaces (essentially showing the phase flip)

Thought for 40s

A magnified view of exactly that region is a great idea: it shows the phase flip itself, not just a label saying it happens. Here's the design, a new panel "MAGNIFIED: THE WAVES AT THE PROBE" under the side view, in both tabs:

- **The layers:** the top glass, the film and the bottom glass, at a scale where one wavelength is about 80–120 px.
- **Four waves, each drawn as an actual wave,** slowly moving along its path:
  1. *The incoming wave*, travelling down through the top glass.
  2. *Wave 1*, reflected at the top of the gap: its crest comes back as a trough when there's a half-wave flip, or as a crest when there isn't. A marker at the interface says which.
  3. *The transmitted wave*, going on down through the film. It's longer there, since $\lambda_{film} = \lambda/n$.
  4. *Wave 2*, reflected at the bottom, back up through the film and the glass, with its own flip marker.
- **The thickness in wavelengths:** "$t = 6.20\ \lambda_{film}$", with $2nt = 12.41\ \lambda$. Since whole wavelengths change nothing, a thick film is drawn with a break that leaves them out ("6 whole wavelengths not drawn: they change nothing"). Only the fraction that matters is shown.
- **At the top,** waves 1 and 2 leave side by side, so students can see whether crest meets crest, with the result spelled out: delay 12.41 + ½ (flip) = 12.91 → 0.91 of a cycle → nearly in step, bright.

Every wave's phase is computed along its own path, so the picture is exact, not a sketch. Let me check that the delay between the two outgoing waves at the top equals 2nt/λ plus the flip:



Ran 6 commands, read 3 files, shared files

The magnified view is in, in both tabs, right under the side view: "MAGNIFIED: THE WAVES AT THE PROBE". It's updated in `po-thin-film-wedge-newton-rings.html`, its MOM version, and `sims-mobile.zip`.

What it shows:

- **Four waves, each drawn as an actual wave,** moving slowly along its path through the glass, the film and the glass below:
    1. *The incoming wave*, travelling down.
    2. *Wave 1*, reflected at the top of the gap.
    3. *The wave going into the film*, visibly longer in an air gap, since λ_film = λ/n.
    4. *Wave 2*, reflected at the bottom and travelling back up through the film and the glass.
- **The phase flips, where they happen:** a marker at each surface says "flipped half a cycle" or "no flip". At a flip you can see the incoming crest come back as a trough.
- **Which wave leaves first:** a dashed line at the top carries a dot on waves 1 and 2 at the same height. On the same side means in step; on opposite sides means opposed.

Beside it, an explanation in plain steps, with the student's own numbers:

1. The wavelength in the film, λ/n.
2. The thickness in film wavelengths, for example "t = 2.900 μm = 5.00 film wavelengths".
3. The round trip: 2t in film wavelengths, which is 2nt/λ wavelengths of the light.
4. The flip: one half-wave flip adds ½ cycle (with which surface it's at).
5. The total delay, for example 10.00 + 0.50 = 10.50 cycles, and "whole cycles change nothing: what counts is 0.50".
6. The verdict: in step and bright, or opposed and dark.

For thick films, the drawing leaves out the whole film wavelengths, marked with a break and the note that "they change nothing". Only the fraction that decides the colour is drawn, so the picture stays readable at any thickness without losing anything physically. I checked that this leaves the result exactly unchanged.

A good case for students to try: a film denser than the glass (n = 1.65). The flip moves from the bottom surface to the top, yet there's still exactly one, so the fringes stay the same.

On a phone, the explanation moves below the drawing and stays readable, as real text at about 13 px rather than text shrunk inside the canvas. Both tabs work on a 360 px phone, and two copies on one MOM page each explain their own film.

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That is a good start. The animation is good but difficult to follow. may be arrows for each of the waves instead of the animated wave with small waves with the appropriate phases right where they change media . The important thing is to visually demonstrate what is happening. May be smaller representative static cycles instead of the ones you made will prevent some from thinking that they correspond to the actual wavelength. For wave two, instead of blue waves arrows down and up labeled as path in the film or path difference. I also like more focus on the reflected view versus transmitted view. The diagram now addresses only the reflected. May be a toggle for either view and each view focusing only on that view.

Thought for 51s

Those are all good teaching calls. The animation shows that waves move, but makes it hard to see the phase, which is the whole point. And full-size wave drawings invite students to think they're seeing the real wavelength. Here's the redesign:

A static diagram of arrows (rays) for each wave, with **small phase symbols** placed right where something happens:

- **Each symbol is a tiny one-cycle wave in a box,** with a dot where it starts, so a crest-first and a trough-first wave look plainly different. A note says they show phase only, not the wavelength.
- **At each reflection, a pair:** "arriving" and "reflected", either the same shape or flipped.
- **On leaving, a comparison box:** wave 1 and wave 2 stacked, so in step or opposed shows at a glance.
- **Wave 2's trip through the film** is shown as down and up arrows labelled "down t + up t = 2t", the path difference, instead of a drawn wave.

A Reflected view / Transmitted view toggle, each focused only on its own light:

- **Reflected:** wave 1 from the top of the gap, wave 2 from the bottom, with one flip between them.
- **Transmitted:** wave 1 goes straight through, and wave 2 is reflected twice (at the bottom, then the top) before leaving. That's two flips, which cancel. It also shows why the transmitted fringes are faint: wave 2 has lost most of its strength in two reflections, keeping only about 4 % of its amplitude. That's the 8.5 % contrast.
- **What follows the toggle:** the two-waves panel, the explanation text and the narration, and the selected pattern band is highlighted.

Let me check the transmitted view's phases before drawing them:

Ran 2 commands

At this dark fringe the phases are as expected: the reflected delay is 0.5 (dark) and the transmitted delay is 0 (bright). No errors, and the animation is gone. Here are both views:



Read 5 files, ran 4 commands, shared files

The magnified view is redesigned along your lines, in `po-thin-film-wedge-newton-rings.html` and its MOM version, updated in `sims-mobile.zip`.

Static arrows instead of animated waves:

- **Each wave is an arrow along its path:** the incoming light, wave 1, and wave 2.
- **Wave 2's trip in the film** is shown as a down arrow and an up arrow labelled "down t" and "up t", with "extra path for wave 2: 2t = 10.00 film wavelengths".
- **Each surface has a marker:** "flip ½" or "no flip".

Small symbols for the phase, not the wavelength:

- **Each is a tiny one-cycle wave in a box,** with a dot marking where it starts, so a crest-first wave and a trough-first wave look plainly different.
- **At each reflection, a pair:** "arriving" and "flipped" or "unchanged", placed right where the light meets the surface.
- **On leaving, a comparison box:** wave 1 and wave 2 stacked, with the verdict ("opposed: DARK", "in step: BRIGHT").
- **The text panel says it plainly:** these symbols show only the phase, not the wavelength. The film is marked "thickness not to scale".

A reflected view / transmitted view toggle, each focused only on its own light:

- **Reflected:** wave 1 reflects at the top of the gap, wave 2 at its bottom, and exactly one of them is flipped.
- **Transmitted:** wave 1 goes straight through, while wave 2 is reflected twice inside the film. Its two flips cancel, so the transmitted pattern is always the opposite of the reflected one. Being reflected twice also leaves wave 2 with only about 4 % of wave 1's amplitude, which is why these fringes are faint. The text says so, and the waves panel draws wave 2 weaker, marked "drawn larger than it is".
- **What follows the toggle:** the explanation, the two-waves panel and its title, the narration, and a highlight on the matching pattern (the band in the wedge tab, the ring image in the Newton tab).

A good case for students to try: a film denser than the glass. The flip moves from wave 2 to wave 1, yet the fringes are unchanged, because there's still exactly one flip between them.

Checked: both tabs in both views work on a 360 px phone without errors, and two copies on one MOM page keep their own views. Their radio buttons stay independent, which I checked specifically, since radio buttons with the same name on one page would interfere.

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